

CARLOS MIGUEL RIVERA SANTOS

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PROFESSIONAL SUMMARY

Results-driven Research Engineer with 4 years of experience in machine learning, data engineering, and cloud infrastructure. Passionate about designing scalable solutions for complex data problems and translating research concepts into production-ready systems. Currently based in Barcelona with a strong track record of delivering impactful projects in fast-paced environments. Seeking challenging opportunities to leverage expertise in modern AI/ML frameworks and distributed systems.

WORK EXPERIENCE

Research Engineer

Nextgen Analytics Solutions (Previously known as DataFlow Insights Inc.)

Barcelona, Spain

June 2022 - Present

<https://www.nextgenanalytics.com>

Architected and deployed machine learning pipelines processing 500GB+ daily data using Python, TensorFlow, and Apache Spark for real-time recommendation systems serving 2M+ users across EMEA region

Implemented advanced LLM-based text analysis features utilizing LangChain and GPT-4 integration, reducing manual content moderation workload by 65% across client portfolio

Led cross-functional team of 3 junior engineers in designing MLOps infrastructure with comprehensive monitoring, versioning, and automated retraining workflows

Published 2 peer-reviewed papers on federated learning optimization in top-tier conferences while maintaining full-time responsibilities in production systems

Senior Data Engineer

TechCore Europe SL

Barcelona, Spain

November 2020 - May 2022

<https://www.techcore.es>

Redesigned data warehouse architecture migrating from legacy systems to cloud-native infrastructure on AWS, reducing query latency by 72% and infrastructure costs by 43%

Built automated ETL pipelines using dbt, Apache Airflow, and Python that processed 200M+ transactions daily with 99.9% reliability SLA

Implemented GitHub Actions CI/CD workflows for data quality validation and automated model deployment, enabling 15+ releases per week

Mentored 2 junior data engineers on SQL optimization, Python best practices, and cloud platform fundamentals

Data Engineer

Axiom Systems (Previously known as CloudData Technologies)

Madrid, Spain

August 2019 - October 2020

<https://axiom-systems.net>

Developed Python-based data pipelines integrating 12+ disparate data sources using Apache Kafka and PostgreSQL, consolidating previously siloed datasets

Implemented monitoring dashboards and alerting systems using Prometheus and Grafana to track pipeline health and data quality metrics

Optimized SQL queries resulting in 8x performance improvement for critical reporting workloads used by 50+ business stakeholders

Collaborated with product and analytics teams to define data schemas and ensure data governance compliance across organization

EDUCATION

Bachelor of Science in Computer Science

Universitat Autònoma de Barcelona (UAB)

Bellaterra, Spain

September 2015 - June 2019

Relevant Coursework: Distributed Systems, Machine Learning, Database Design, Software Engineering

TECHNICAL SKILLS

Programming Languages: Python, SQL, Scala, JavaScript, Bash

Data Engineering: Apache Spark, Apache Kafka, Apache Airflow, dbt, ETL/ELT Pipelines

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, LangChain, Hugging Face Transformers, LLMs, Feature Engineering

Cloud Platforms: AWS (EC2, S3, RDS, Lambda, SageMaker), Google Cloud Platform (BigQuery, Dataflow)

Container & Orchestration: Docker, Kubernetes, Terraform, ArgoCD

Databases: PostgreSQL, MySQL, MongoDB, Elasticsearch

DevOps & Tools: Git, GitHub Actions, Prometheus, Grafana, Jenkins, Linux

Analytics & Visualization: Tableau, Looker, Apache Superset

LANGUAGES

Spanish: Native

English: Fluent (C1)

Catalan: Fluent

CERTIFICATIONS

AWS Certified Solutions Architect - Associate (2021)

Machine Learning Engineer Specialization - Coursera (2020)